Ian Douglas Mager

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SUMMARY

Entry-level Software Engineer with hands-on experience across full-stack development, robotics, and automated testing. Comfortable working independently or in a team, with a track record of shipping complete projects from design through deployment.

EDUCATION

Texas State University – Bachelor of Science in Computer Science August 2022 – May 2026

San Marcos, TX | Relevant Coursework: Software Engineering, Machine Learning, Compiler Construction, Database Systems, Robotics, Algorithms & Data Structures

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL

Frameworks & Tools: FastAPI, PostgreSQL, Supabase, Firebase, Node.js, React, Next.js, Git, Docker, GitHub Actions, Jira, Bitbucket, scikit-learn, Jest

EXPERIENCE

Cashier | H-E-B 2021 – Aug 2022

- Provided fast, accurate customer service in a high-volume retail environment while balancing coursework, developing reliability and time-management skills carried into technical project work.

PROJECTS

Autonomous Robot Navigation System | Texas State University Jan 2026 – May 2026

- Collaborated with a 4-person team to design and implement an autonomous navigation system for a TurtleBot3 platform using ROS 2, integrating camera and LiDAR sensor data to enable real-time environment perception.
- Developed and tuned navigation logic responsible for goal-directed path planning, enabling the robot to autonomously interpret a target destination and execute collision-free movement to reach it.
- Delivered a fully functional robot capable of completing all assigned navigation tasks in an end-of-semester demonstration, validating the accuracy and reliability of the sensor-fusion and path-planning pipeline.

IntelliPlay – AI-Powered Daily Puzzle Application | Team Project 2025

- Collaborated with a 3-person team (using Jira for sprint planning and Bitbucket for version control) to design and build a full-stack puzzle application serving unique, AI-generated visual puzzles on a daily rotation.
- Engineered the core puzzle generation pipeline in JavaScript, integrating DALL·E for image synthesis and Gemini for dynamic content generation, with Firebase handling authentication, storage, and real-time data.
- Built out a Jest/Babel unit testing suite to validate puzzle generation and scoring logic, and deployed the application on Render with an automated daily rotation mechanism.

Machine Learning – Kaggle Competition Project | Texas State University Spring 2026

- Led model selection and feature engineering for a multi-class irrigation prediction task using scikit-learn, applying cross-validation and regularization strategies to address class imbalance on real-world agricultural data.